

Hybrid Hierarchical Deep Learning Framework for Stock Forecasting in the Taiwan Stock Market

Abstract

With the advancement of financial technology, deep learning has shown strong potential in financial time-series forecasting. This study proposes a hybrid hierarchical framework that integrates a hierarchical feature selection system (HFSLS), particle swarm optimization (PSO), and bidirectional gated recurrent units (BIGRU), and evaluates its robustness in the Taiwan stock market.

The proposed model reduces feature redundancy through HFSLS, applies PSO for hyperparameter tuning, and captures temporal dependencies using BIGRU. Representative stocks from different industries are selected to assess model performance under diverse market conditions.

Experimental results indicate that feature selection plays a critical role in improving predictive stability, while PSO achieves better performance when combined with feature reduction. Overall, the proposed framework demonstrates consistent performance improvements over baseline models, highlighting its effectiveness for quantitative investment and back testing applications.

Keywords: Financial Time-Series Forecasting, HFSLS, PSO, BIGRU, Feature Selection, Quantitative Investment, Taiwan Stock Market

1 Introduction

1.1 Research Background

With the rapid development of financial technology, machine learning and deep learning have become increasingly important in stock price prediction. Traditional approaches, including technical analysis and econometric models, typically rely on a limited number of indicators and linear assumptions, which restrict their ability to capture complex and nonlinear market dynamics (Fama & French, 1992; Malkiel, 2003).

In recent years, researchers have increasingly incorporated high-dimensional features and advanced learning algorithms to enhance predictive performance. Empirical studies have demonstrated that machine learning methods significantly improve asset return predictability compared to traditional models (Gu et al., 2020). In particular, deep learning architectures are capable of capturing nonlinear relationships and temporal dependencies in financial data (LeCun et al., 2015).

Among these approaches, recurrent neural networks and their variants have been widely adopted for time series forecasting. The Long Short-Term Memory (LSTM) model addresses the vanishing gradient problem inherent in traditional recurrent neural networks (Hochreiter & Schmidhuber, 1997; Bengio et al., 1994), while the Gated Recurrent Unit (GRU) provides a more computationally efficient alternative (Cho et al., 2014). Furthermore, bidirectional recurrent structures enhance predictive performance by incorporating both past and future contextual information (Schuster & Paliwal, 1997).

1.2 Research Motivation

However, when handling high-dimensional feature sets, models are prone to the curse of dimensionality and overfitting, which can significantly degrade out-of-sample performance (Guyon & Elisseeff, 2003; Hastie et al., 2009). In financial applications, this issue is particularly critical, as improper model validation may lead to severe backtest overfitting (Bailey et al., 2014; López de Prado, 2018).

To address these challenges, this study introduces a hierarchical feature selection mechanism to identify informative features and reduce redundancy, supported by

theoretical and empirical findings in feature selection research (Li et al., 2017). In addition, particle swarm optimization is employed to optimize model hyperparameters efficiently (Kennedy & Eberhart, 1995; Bergstra & Bengio, 2012), thereby improving model stability and predictive performance.

Furthermore, applying a model originally designed for the Chinese stock market to the Taiwan stock market enables evaluation under different market structures, regulatory environments, and investor behaviors. This cross-market validation provides a more rigorous assessment of model robustness and generalization.

1.3 Research Objectives

The objective of this study is to develop a unified hierarchical framework for stock price prediction that integrates feature selection, hyperparameter optimization, and deep learning-based sequence modeling. The proposed framework is designed to address the challenges associated with high-dimensional financial data, particularly feature redundancy and overfitting, while improving predictive accuracy, stability, and out-of-sample generalization performance.

In addition, this study evaluates the robustness and transferability of the proposed framework by applying it to the Taiwan stock market, which differs from the original market in terms of structure, regulatory environment, and investor behavior. To further assess its robustness under heterogeneous market conditions, the model is tested on representative stocks across multiple industries.

Finally, systematic ablation experiments are conducted to examine the impact of each component within the proposed framework, providing empirical insight into the effectiveness of feature selection and hyperparameter optimization in improving forecasting performance.

2 Literature Review

2.1 Traditional Statistical Methods

Traditional statistical methods have long been widely used in financial time series forecasting. The Autoregressive Moving Average (ARMA) model was systematically developed by Box and Jenkins (1970), capturing linear dependencies between observations and error terms. This framework was later extended to the Autoregressive Integrated Moving Average (ARIMA) model to handle non-stationary time series (Box & Jenkins, 1976).

To model volatility clustering, a common phenomenon in financial markets, Engle (1982) proposed the Autoregressive Conditional Heteroskedasticity (ARCH) model, which was further generalized into the GARCH model by Bollerslev (1986). These models provide a strong theoretical foundation for modeling time-varying volatility.

However, these approaches rely on assumptions of linearity and stationarity, which are often violated in real financial markets characterized by nonlinear dynamics and structural breaks (Hastie et al., 2009). As a result, their predictive performance is often limited in complex, high-dimensional environments.

2.2 Machine Learning Methods

To overcome the limitations of traditional models, machine learning techniques have been increasingly applied to financial forecasting. Methods such as Support Vector Machines (Vapnik, 1995), Random Forests (Breiman, 2001), and other nonparametric approaches can capture nonlinear relationships and interactions among variables.

Despite their flexibility, machine learning models are still susceptible to overfitting, particularly in high-dimensional settings (Hastie et al., 2009; Guyon & Elisseeff, 2003). Moreover, many traditional machine learning models lack the ability to capture temporal dependencies inherent in financial time series.

Recent empirical studies demonstrate that machine learning significantly enhances asset pricing and return prediction (Gu et al., 2020; Kelly et al., 2019), highlighting the

importance of integrating advanced modeling techniques with robust feature selection strategies.

2.3 Deep Learning Methods

Deep learning models have shown strong capabilities in modeling complex nonlinear relationships through hierarchical feature representation learning (LeCun et al., 2015). Recurrent Neural Networks (RNNs) are particularly suitable for sequential data; however, they suffer from the vanishing gradient problem, limiting their effectiveness in long-term dependency modeling (Bengio et al., 1994).

The Long Short-Term Memory (LSTM) model addresses this issue through gating mechanisms (Hochreiter & Schmidhuber, 1997), while the Gated Recurrent Unit (GRU) simplifies the architecture and improves computational efficiency (Cho et al., 2014). Bidirectional RNNs further enhance performance by incorporating both forward and backward information (Schuster & Paliwal, 1997).

More recently, attention-based models and Transformer architectures have demonstrated superior performance in capturing long-range dependencies (Vaswani et al., 2017; Lim et al., 2021). However, these models often require large datasets and high computational resources, which may limit their applicability in certain financial contexts.

Despite their advantages, deep learning models remain sensitive to high-dimensional and redundant inputs, leading to potential overfitting (Goodfellow et al., 2016). Therefore, integrating effective feature selection and hyperparameter optimization techniques is essential for improving model robustness and generalization performance.

3 Methodology

3.1 Data Preprocessing and Normalization

Stock market data contain multi-dimensional features such as technical indicators, and the value ranges of these features vary significantly. To prevent instability during model training, all features are normalized to the range $[-1, 1]$ to accelerate convergence and improve overall model performance. The normalization formula is:

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where x represents the feature vector, and x_{min} and x_{max} denote the minimum and maximum values of the feature, respectively.

3.2 Hierarchical Feature Selection with Local Shuffling (HFSLS)

Traditional feature selection methods often require retraining the model to evaluate feature importance, resulting in high computational cost. This study adopts Hierarchical Feature Selection with Local Shuffling (HFSLS), which assesses feature importance by perturbing individual features and measuring the resulting change in prediction loss, without retraining the model.

1. Original prediction error:

$$L = (\bar{M} - y)^2 \quad (2)$$

where $\bar{M}$ is the model's mean prediction on the original data and y is the true label.

2. Perturbed prediction error for feature i :

$$L_i = (\bar{M}_i - y)^2 \quad (3)$$

where $\bar{M}_i$ represents the mean prediction after shuffling feature i .

3. Feature importance evaluation:

$$E_i = \frac{\sigma(L_i - L)}{E(L_i - L)} \quad (4)$$

where E is the expectation operator and σ is the standard deviation. A significant increase in E_i indicates high feature importance.

In the hierarchical framework (Fig 1), each layer selects features from the previous layer using local shuffling and sampling, ensuring that the number of features decreases layer by layer ($|F_1| > |F_2| > \dots > |F_K|$), effectively reducing redundancy and computational complexity.

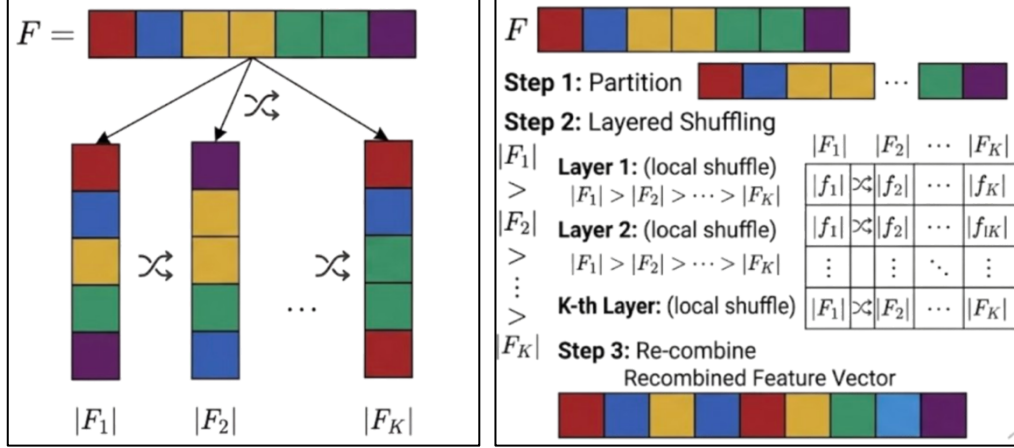


Fig 1. Global shuffling (left) and Local shuffling (right)

3.3 Bidirectional Gated Recurrent Unit (BIGRU)

To capture the temporal dynamics and contextual dependencies in stock price time series, this study employs BIGRU as the core prediction model. GRU (Fig 2) mitigates the vanishing gradient problem of traditional RNNs through Update Gate (z_t) and Reset Gate (r_t) mechanisms.

1. Candidate hidden state:

$$\tilde{h}_t = \tanh(W_{ih}x_t + W_{hh}(r_t \odot h_{t-1}) + b_h) \quad (5)$$

2. Hidden state update:

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (6)$$

BIGRU (Fig 3) combines forward and backward sequences to capture complete temporal context (e.g., yesterday $\rightarrow$ today $\rightarrow$ tomorrow). The final output is the concatenation of forward and backward hidden states:

$$h_t = \text{concat}(h_t^{\text{forward}}, h_t^{\text{backward}}) \quad (7)$$

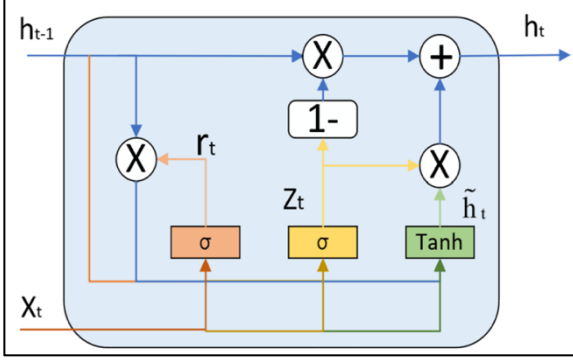


Fig 2. Internal structure diagram of GRU

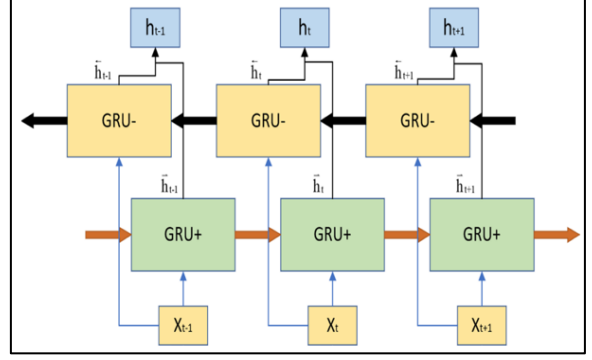


Fig 3. Structure diagram of BIGRU

(Reproduced from Chen and Tang (2025) [21])

3.4 Particle Swarm Optimization (PSO) for Hyperparameter Tuning

The performance of deep neural networks heavily depends on hyperparameter configuration (e.g., GRU units, Dropout rate, learning rate). Manual tuning is inefficient and prone to local optima. This study employs Particle Swarm Optimization (PSO) to automatically search for the global optimal hyperparameters of the BIGRU model.

- Velocity update:

$$v_i(t+1) = \omega v_i(t) + c_1 r_1 (P_i^{\text{best}} - x_i(t)) + c_2 r_2 (P_g^{\text{best}} - x_i(t)) \quad (8)$$

- Inertia weight:

$$\omega = \omega_{\max} - \frac{n}{n_{\max}} (\omega_{\max} - \omega_{\min}) \quad (9)$$

- Position update:

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (10)$$

The fitness function is the Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (11)$$

Particles iteratively update their positions and velocities until reaching the maximum iterations or the global best hyperparameter set.

3.5 Multi-layer Model Integration and Weighted Aggregation

In the proposed HFSLS-PSO-BIGRU hierarchical framework (Fig 4), each layer generates an individual prediction based on progressively refined feature subsets. To obtain a unified prediction, a weighted aggregation mechanism is employed to combine the outputs from all layers.

Let k denote the total number of layers and let $\hat{y}_i$ represent the prediction from the i -th layer. The final prediction is computed as:

$$\hat{y} = \sum_{i=1}^k w_i \cdot \hat{y}_i \quad (12)$$

where w_i is the weight assigned to the i -th layer. The weights are defined as:

$$w_i = \frac{1+(i-1)s}{k+\frac{k(k-1)s}{2}}, i = 1, 2, \dots, k \quad (13)$$

where s is the weighting step parameter controlling the contribution difference between layers. This formulation ensures that all weights are normalized and sum to one.

When $s = 0$, the weighting scheme reduces to uniform averaging, i.e., $w_i = \frac{1}{k}$, indicating equal contribution from all layers. Experimental results show that this setting provides the most stable predictive performance. For $s > 0$, higher layers receive larger weights, reflecting their refined feature representations.

Finally, the aggregated output integrates multi-layer feature selection and PSO-optimized model configurations, producing the prediction with minimized loss and improved generalization capability.

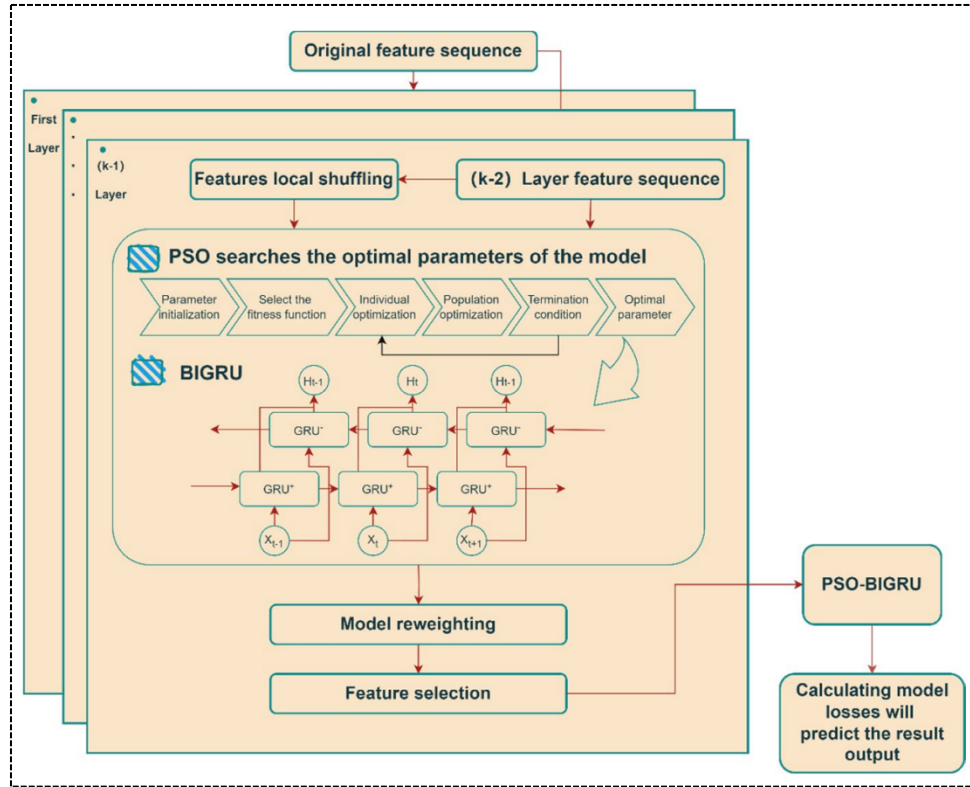


Fig 4. HFSLS-PSO-BIGRU Model Framework

(Reproduced from Chen and Tang (2025) [21])

3.6 Logarithmic Return (Ln Return)

The rate of return used in this study is the logarithmic return, rather than the simple return, because the logarithmic return possesses the property of additivity. It is not affected by the base period or time interval, and the percentage change in both upward and downward movements is symmetric.

The concept is to take the natural logarithm of the ratio between the closing price of the next period and that of the current period, expressed as follows:

$$R_{t+h} = \ln \left(\frac{S_{t+h}}{S_t} \right) \quad (14)$$

where R_{t+h} represents the return over a holding period of h months, S_{t+h} denotes the stock closing price at time $t + h$, and S_t denotes the stock closing price at time t .

4 Experiments

4.1 Data Sources

The dataset used in this study is sourced from the Taiwan stock market and obtained via the TEJ Pro database. The study period spans **from January 1, 2020, to September 2, 2025**, and consists of **daily closing price data**, ensuring both timeliness and practical relevance while capturing complete market cycles. To avoid information leakage, the dataset is partitioned based on chronological order. The first 70% of the data, covering the period from **January 1, 2020, to December 31, 2023**, is used as the training set, while the remaining 30%, spanning from **January 1, 2024, to September 2, 2025**, serves as the test set.

The primary prediction target of this study is the Taiwan Stock Exchange Capitalization Weighted Stock Index (TAIEX), representing overall market trends. To evaluate the model's generalization ability across industries, four representative individual stocks are also included: TSMC (semiconductor), Evergreen Marine (shipping), MediaTek (IC design), and Uni-President Enterprises (retail), covering diverse sectors and enhancing the cross-industry representativeness of the experimental results.

For feature construction, raw price and trading data from TEJ Pro are used to programmatically calculate the Alpha158 feature set, comprising 158 structurally consistent technical indicators spanning price trends, momentum, volatility, and other dimensions. Constructing these features independently, rather than using pre-built indicators, ensures consistency, controllability, and reproducibility.

Given that high-dimensional features may introduce redundancy and overfitting, a hierarchical feature selection with local shuffling (HFSLS) approach is applied. This method performs layered screening and dimensionality reduction to enhance model stability and predictive performance.

In summary, by integrating market indices and multi-industry individual stocks with high-dimensional technical indicators and a structured feature selection mechanism,

this study aims to thoroughly evaluate the robustness and cross-market generalization ability of the proposed HFSLS-PSO-BIGRU model.

4.2 Feature Engineering and Preprocessing

4.2.1 Alpha158 Feature Set Construction

The Alpha158 feature set is employed as the model input, comprising 158 technical indicators with significant economic implications. These indicators span multiple dimensions to capture short, medium, and long-term market dynamics:

- **Basic Price-Volume Indicators:** Includes daily open, high, low, and close (OHLC) prices, trading volume, and the Volume Weighted Average Price (VWAP).
- **Trend Indicators:** Leverages Moving Averages (MA) of various periods (5, 10, 20, 30, and 60 days) along with K-line pattern features.
- **Momentum Indicators:** Includes the Rate of Change (ROC) and the Raw Stochastic Value (RSV) to measure price momentum.
- **Volatility and Correlation:** Utilizes standard deviation, range extremes, and the Beta coefficient to quantify market risk and relative strength.

4.2.2 Data Preprocessing Workflow

1. **Missing Value Imputation:** For minor missing data points caused by unexpected trading halts, linear interpolation is applied to maintain the continuity of the time-series data.
2. **Sliding Window Processing:** To accommodate the temporal characteristics of the Bidirectional Gated Recurrent Unit (BIGRU), a sliding window technique is used for sample segmentation. This transforms historical data into sequential feature inputs, allowing the model to capture dynamic temporal dependencies effectively.

4.3 Evaluation Metrics

To comprehensively evaluate the predictive performance of the proposed model, three widely adopted regression metrics are employed: Root Mean Squared Error (RMSE),

Mean Absolute Error (MAE), and the coefficient of determination (R^2). These metrics provide complementary perspectives on model performance, including error magnitude, robustness to outliers, and explanatory power.

Root Mean Squared Error (RMSE):

RMSE measures the square root of the average squared differences between predicted values and actual values. It is derived from the Mean Squared Error (MSE) by taking the square root, thereby restoring the original unit of the target variable and improving interpretability. Due to the squaring of errors, RMSE assigns greater penalties to larger deviations and is therefore more sensitive to outliers. The formula is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (15)$$

Mean Absolute Error (MAE):

MAE calculates the average of the absolute differences between predicted values and actual values. Unlike RMSE, MAE treats all errors equally and does not disproportionately emphasize large deviations, making it more robust to outliers. Additionally, MAE is directly interpretable, as it retains the same unit as the original data. The formula is given by:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (16)$$

Coefficient of Determination (R^2):

The coefficient of determination, denoted as R^2 , evaluates the proportion of variance in the dependent variable that can be explained by the model. It reflects the goodness-of-fit of the model, with values typically ranging from 0 to 1. A higher R^2 indicates that the model better captures the underlying data patterns. The formula is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (17)$$

where y_i represents the actual value, $\hat{y}_i$ denotes the predicted value, and $\bar{y}$ is the mean of the actual values.

4.4 Model Configuration and Hyperparameters

Table 1. Hyperparameter Configurations for HFSLs and BIGRU

Category	Parameter	Value
HFSLs	Sub-models (K)	3
	Selection Ratios	[1.0, 0.9, 0.8]
	Sampling Bins	3
	Window Size	30
BIGRU	Optimizer	Adam
	Loss Function	MSE
	Hidden Layers	Bidirectional
	Activation	ReLU
	Training Epochs	30
	Batch Size	32

Table 2. PSO Optimization Search Space and Swarm Parameters

Type	Parameter	Value
Search Space	Hidden Units	[32, 96]
	Dropout Rate	[0.1, 0.4]
	Learning Rate	[0.0005, 0.005]
Swarm Config	Number of Particles	10
	Maximum Iterations	25
	Inertia Weight (ω)	0.7
	Cognitive Coefficient (C1)	1.5
	Social Coefficient (C2)	1.5

5. Research Results

5.1 Taiwan Stock Exchange Weighted Index (TAIEX)

Table 3. Ablation Study Results for TAIEX Prediction

Model	R ²	MSE	RMSE	MAE	MAPE	MAE-Original
BIGRU	-1.457	0.069	0.264	0.252	7.993	1,784.400
PSO-BIGRU	-3.118	0.117	0.341	0.329	10.451	2,331.433
HFSLS-BIGRU	0.935	0.002	0.043	0.030	0.985	215.344
HFSLS-PSO-BIGRU	0.937	0.002	0.042	0.032	1.040	226.928

Note: MAE-Original represents the error rescaled back to the original units to provide practical interpretability.

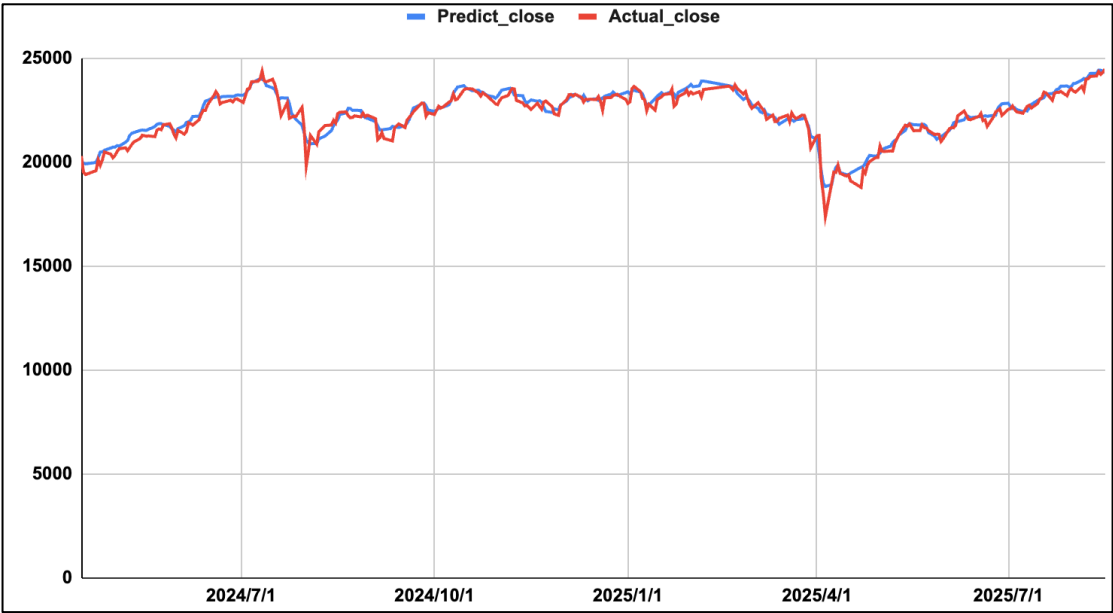


Fig 5. TAIEX Prediction Results Using the HFSLS-PSO-BIGRU model

As shown in Fig 5, the predicted values align closely with the actual TAIEX closing prices, particularly during high-volatility periods between 2024 and 2025. Although a slight lag occurs during the sharp market correction in April 2025, the overall predictive trajectory remains highly stable.

Coupled with the quantitative results in Table 3, this visualization confirms that the HFSLS module successfully filters out market noise and retains critical features. Consequently, the proposed framework demonstrates exceptional generalization capability and predictive stability over extended time series.

5.2 TSMC, Evergreen, MediaTek, and Uni-President

Table 4. Performance Comparison of Model Generalization Across Different Stocks

Evaluation Index	Stock	BIGRU	PSO-BIGRU	HFSLS-BIGRU	HFSLS-PSO-BIGRU
R ²	TSMC	-0.703	-3.893	0.952	0.959
	Evergreen	0.266	0.115	0.894	0.731
	MediaTek	0.544	-3.101	0.905	0.928
	Uni-President	0.837	0.608	0.945	0.953
MSE	TSMC	0.084	0.243	0.002	0.002
	Evergreen	0.033	0.008	0.005	0.012
	MediaTek	0.02	0.174	0.004	0.003
	Uni-President	0.005	0.013	0.001	0.001
RMSE	TSMC	0.29	0.493	0.049	0.045
	Evergreen	0.183	0.09	0.068	0.109
	MediaTek	0.142	0.417	0.063	0.055
	Uni-President	0.072	0.114	0.034	0.039
MAE	TSMC	0.273	0.476	0.039	0.035
	Evergreen	0.15	0.07	0.049	0.088
	MediaTek	0.107	0.4	0.048	0.043
	Uni-President	0.055	0.096	0.025	0.028
MAPE	TSMC	12.354	21.543	1.838	1.64
	Evergreen	7.071	3.486	2.385	4.256
	MediaTek	4.918	17.849	2.179	1.953
	Uni-President	1.358	2.355	0.62	0.703
MAE-Original	TSMC	124.873	218.034	18.036	16.17
	Evergreen	15.012	7.031	4.884	8.774
	MediaTek	63.653	237.734	28.789	25.803
	Uni-President	3.715	6.42	1.67	1.902

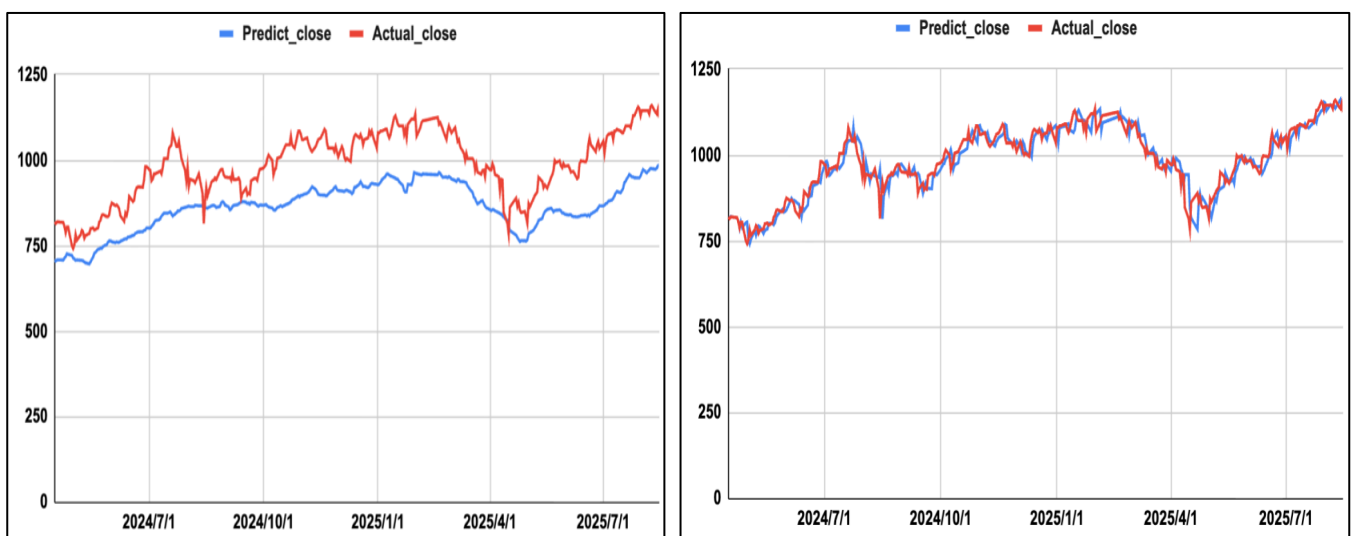


Fig 6. TSMC Baseline (left) vs. Proposed Model (right)

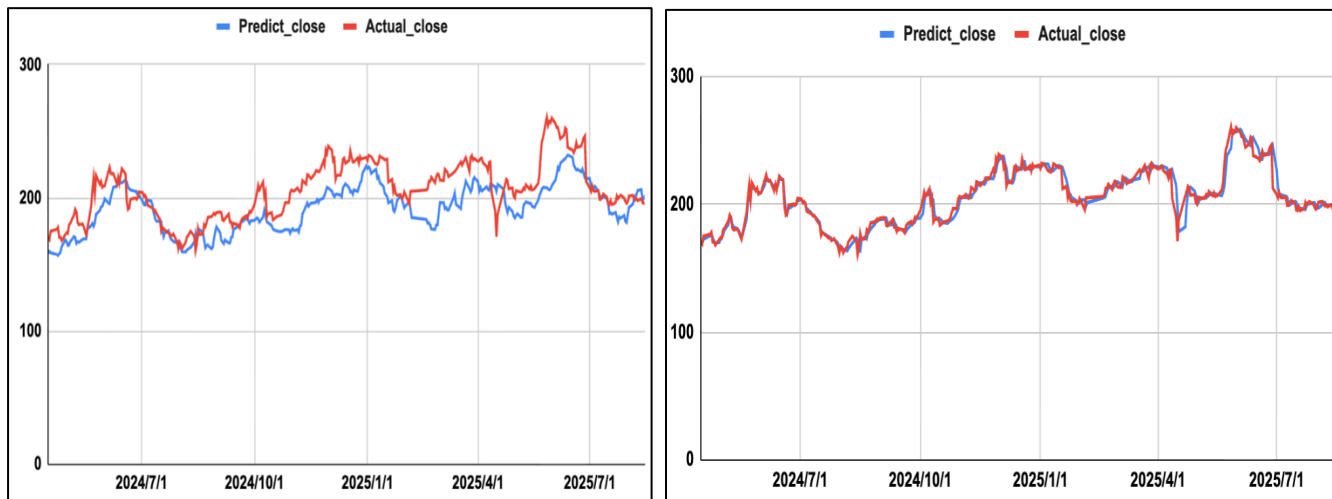


Fig 7. Evergreen Baseline (left) vs. Proposed Model (right)

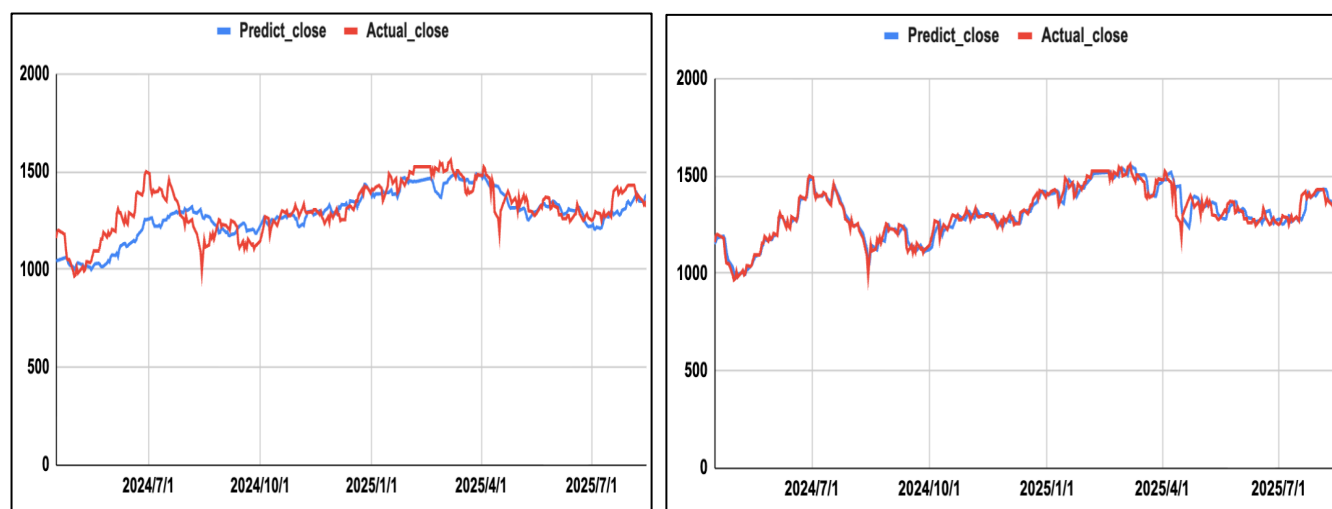


Fig 8. MediaTek Baseline (left) vs. Proposed Model (right)

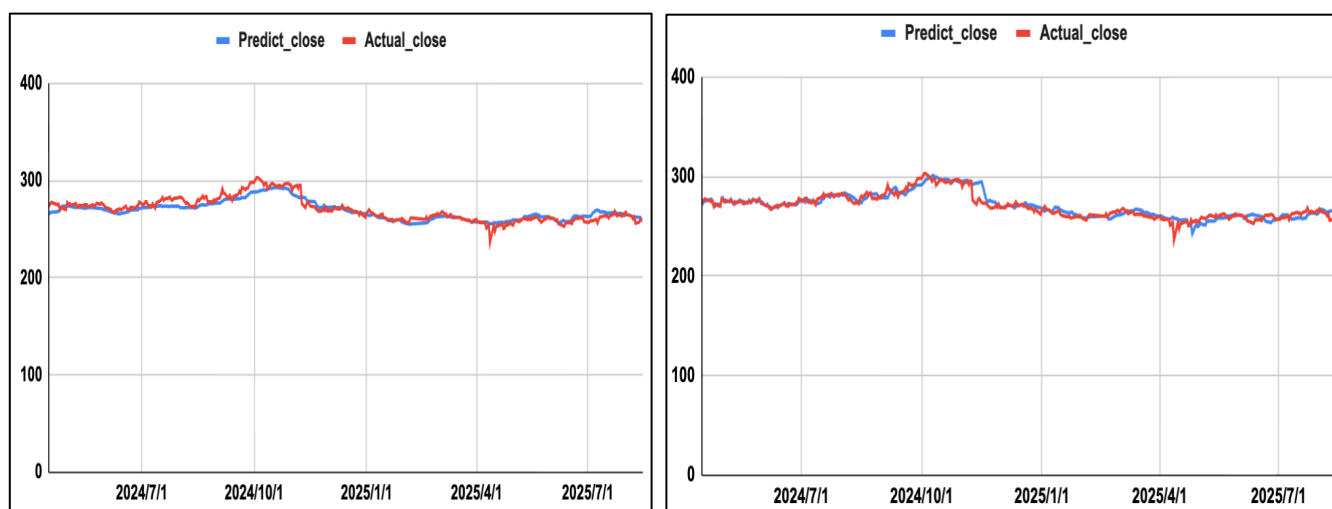


Fig 9. Uni-President Baseline (left) vs. Proposed Model (right)

5.3 Ablation Study Analysis

To systematically evaluate the contribution of each component to the prediction of the Taiwan Stock Exchange Weighted Index (TAIEX), an ablation study is conducted by comparing different model configurations.

The results show that HFSLs-BIGRU significantly outperforms the baseline BIGRU model, with an improvement of 2.392 in R^2 and a 97.1% reduction in MSE. This indicates that the HFSLs module plays a critical role in enhancing predictive performance. In contrast, PSO-BIGRU performs worse than the baseline model, with a decrease of 1.661 in R^2 and an increase of 69.6% in MSE, suggesting that applying PSO alone in a high-dimensional feature space may negatively affect model performance.

Further comparison between the full model and HFSLs-BIGRU reveals only a marginal improvement of 0.002 in R^2 , with MSE remaining nearly unchanged. This suggests that the additional contribution of PSO when combined with HFSLs is limited. Overall, the full model achieves a total improvement of 2.394 in R^2 and a 97.1% reduction in MSE compared to the baseline BIGRU model.

These findings indicate that HFSLs is the primary contributor to performance improvement, accounting for many of the observed gains. On the other hand, PSO may introduce instability when applied independently, and its synergistic effect with HFSLs appears to be minimal, with performance gains of less than 0.3% compared to using HFSLs alone.

In terms of cross-stock stability, the baseline BIGRU model achieves an average R^2 of only 0.088 with a high standard deviation of 0.656, indicating unstable performance across different stocks. The PSO-BIGRU model exhibits an average R^2 of -0.768 and an even larger standard deviation of 2.045, reflecting severe instability. In contrast, HFSLs-BIGRU achieves the best performance with an average R^2 of 0.926 and a low standard deviation of 0.029, demonstrating both high accuracy and strong stability. The full model attains an average R^2 of 0.902 with a standard deviation of 0.096, and achieves the best performance in four out of five prediction targets, indicating robust generalization ability.

5.4 Robustness Evaluation and Cross-Sector Analysis

To evaluate the generalization capability of the HFSL-PSO-BIGRU framework, this section compares the baseline model (left panels) and the proposed model (right panels) across four representative stocks (Fig 6–9).

For TSMC (Fig 6), the baseline model exhibits clear prediction drift, failing to fully capture the 2024 upward trend. In contrast, the proposed model closely follows the underlying price trajectory, indicating improved temporal alignment.

For Evergreen (Fig 7) and MediaTek (Fig 8), which are characterized by high volatility, the baseline model shows unstable responses to abrupt price fluctuations. The proposed model, however, maintains more consistent tracking performance under rapid market changes, suggesting improved robustness in nonlinear and non-stationary conditions.

For Uni-President (Fig 9), representing a relatively low-volatility scenario, the baseline model still presents noticeable oscillations around the ground truth. In comparison, the proposed model produces a smoother and more stable predictive path.

Overall, the visual comparisons in Fig 6–9 demonstrate that the proposed framework achieves more consistent alignment with actual price movements across different market regimes, highlighting its robustness and generalization capability in financial time-series forecasting.

6. Trading Strategy Design

6.1 Overlapping Trade Problem

The proposed model in this study is designed to predict the price movement at $T + 11$. In practical implementation, if the model generates a new prediction signal on each trading day and trading decisions are executed accordingly, an “overlapping trade problem” arises.

Specifically, a position initiated at time t is held until $t + 11$, while a new prediction generated at $t + 1$ may trigger another trade. As illustrated in Fig 10, multiple positions with different entry times but overlapping holding periods may coexist within the portfolio at any given time. This phenomenon increases the complexity of interpreting backtesting results and may affect the accuracy of performance evaluation.

Date	Trading Status
Day1	Open Position (Stock A) 
Day2	Open Position (Stock B) 
Day3	Open Position (Stock C) 
Day11	Open Position (Stock K) 

Fig 10. Illustration of the Overlapping Trade Problem

6.2 Evaluation of Trading Schemes

To address the overlapping trade problem, two trading schemes are considered in this study: the non-overlapping trading method and the capital allocation method.

1. Non-Overlapping Trading Method

This method follows a sequential “open–hold–close” cycle. Once a position is initiated at time t , no new trades are executed until the position is closed after 11 trading days, i.e., during the period from $t + 1$ to $t + 10$. As illustrated in Fig 11, each trade is

executed only after the previous position has been fully closed, thereby eliminating overlapping positions.

Although this method completely avoids overlapping trades, it fails to fully utilize the daily prediction signals generated by the model. During the testing period (January 2024 to August 2025), each stock provides 286 valid prediction opportunities. However, under this method, only approximately $\frac{286}{11}$ trades can be executed, resulting in a limited sample size, reduced statistical reliability, and inefficient capital utilization.

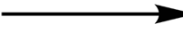
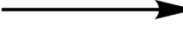
Date	Trading Status
Day1	Open Position (Stock A)  Close Position (Stock A)
Day2-11	
Day12	Open Position (Stock B)  Close Position (Stock B)
Day13-22	
Day23	Open Position (Stock C)  Close Position (Stock C)

Fig 11. Illustration of the Non-Overlapping Trading Method

2. Capital Allocation Method

The capital allocation method is adopted as the primary trading strategy in this study. This approach can also be interpreted as a sub-portfolio aggregation framework.

Specifically, the initial capital is evenly divided into 11 equal portions, each representing an independent sub-portfolio. On each trading day, a trading decision is made based on the model’s prediction signal, and one sub-portfolio is used to initiate a position, which is then held for 11 trading days before being closed. As illustrated in Fig 12, this design enables continuous execution of daily trading signals without capital overlap.

Under this framework, the portfolio can hold up to 11 positions simultaneously, each with a different entry time, thereby forming a time-diversified trading mechanism. The overall portfolio return is calculated as the average return of all individual trades:

$$R_{\text{portfolio}} = \frac{1}{N} \sum_{i=1}^N R_i \quad (18)$$

where R_i represents the return of the i -th trade and N denotes the total number of trades.

Day 1	Day 2	Day 3	Day 4	Day 5	...	Day 11
$\frac{\$1,000,000}{11}$	$\frac{\$1,000,000}{11}$	$\frac{\$1,000,000}{11}$	$\frac{\$1,000,000}{11}$	$\frac{\$1,000,000}{11}$	...	$\frac{\$1,000,000}{11}$

Fig 12. Illustration of the Capital Allocation Method

Note: The total initial capital is set to \$1,000,000 and evenly divided into 11 sub-portfolios.

6.3 Back testing Methodology

The back testing period in this study is set from **January 1, 2024, to August 18, 2025**. During this period, each target stock contains 393 valid trading observations, ensuring a sufficient sample size for statistical representativeness.

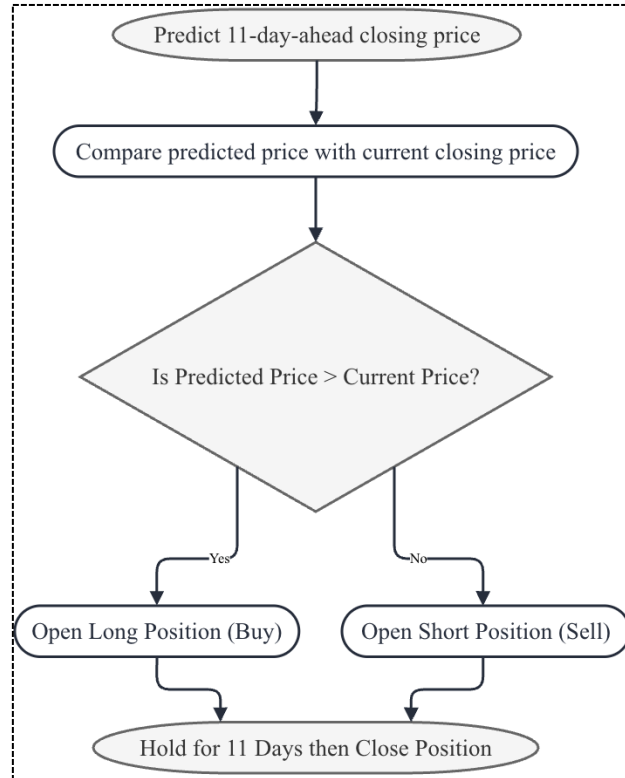


Fig 13. Decision-making flowchart of the proposed long-short trading strategy

The trading logic is illustrated in Fig 13. A symmetric long–short strategy is adopted in this study. Specifically, the model predicts the closing price of a target stock at $T + 11$, which is then compared with the current closing price. If the predicted price is higher than the current price, a bullish signal is generated, and a long position is initiated. Conversely, if the predicted price is lower than the current price, a short position is taken. All positions are held for a fixed period of 11 trading days and are closed at the end of the holding period, ensuring consistency between the prediction horizon and the trading strategy.

7 Model Results Analysis

7.1 BIGRU Model

Table 5. Back Testing Performance Metrics of the BIGRU Model

Performance Metric	TSMC	Evergreen	MediaTek	Uni-President
Total Trades	383	383	383	383
Long Trades	42	95	174	154
Short Trades	341	288	209	229
Long Win Rate (%)	100	77.895	81.609	80.519
Short Win Rate (%)	35.191	54.861	61.722	71.616
Total Win Rate (%)	42.298	60.574	70.757	75.196
Strategy Return (%)	-23.571	107.925	217.962	27.074
Buy-and-Hold Return (%)	101.195	26.77	46.911	-4.003
Excess Return (%)	-124.765	81.154	171.052	31.077
Average Return Per Trade (%)	-0.769	2.125	3.378	0.691
Profit Factor	0.746	2.025	3.448	2.767

Note: Excess Return is calculated as the difference between the Strategy Return and the Buy-and-Hold Return. Profit Factor is defined as the ratio of total profits to total losses.

Table 5 presents the trading performance of the baseline BIGRU model during the testing period across four target stocks. A total of 383 trades were executed for each stock, with long trades representing a smaller portion for TSMC (42 trades) and a higher proportion for MediaTek and Uni-President (174 and 154 trades, respectively). Short trades dominate TSMC (341 trades), indicating different operational tendencies across stocks.

Total win rates vary considerably: TSMC achieves only 42.30%, while MediaTek and Uni-President reach 70.76% and 75.20%, respectively. Long trade win rates generally

exceed short trade win rates; however, TSMC's high long win rate (100%) is offset by a low short win rate, limiting overall performance.

Strategy returns further highlight performance differences: TSMC yields -23.57%, well below the buy-and-hold return (101.20%), with an average return per trade of -0.77%. Profit factors show TSMC at 0.75 and MediaTek at 3.45, suggesting some profitable trades but limited overall robustness. Overall, the baseline BIGRU model exhibits weak trading performance during the testing period, with both stability and return potential remaining constrained.

7.2 HFSLs-PSO-BIGRU Model

Table 6. Back Testing Performance Metrics of the HFSLs-PSO-BIGRU Model

Performance Metric	TSMC	Evergreen	MediaTek	Uni-President
Total Trades	383	383	383	383
Long Trades	247	182	195	174
Short Trades	146	211	198	219
Long Win Rate (%)	93.927	95.604	96.923	87.931
Short Win Rate (%)	78.767	84.834	78.283	84.475
Total Win Rate (%)	88.295	89.821	87.532	86.005
Strategy Return (%)	356.857	598.816	563.712	42.502
Buy-and-Hold Return (%)	99.252	21.74	48.389	-5.328
Excess Return (%)	257.605	577.076	515.323	47.83
Average Return Per Trade (%)	4.344	5.59	5.44	0.996
Profit Factor	20.448	29.532	25.208	6.6

Note: Excess Return is calculated as the difference between the Strategy Return and the Buy-and-Hold Return. Profit Factor is defined as the ratio of total profits to total losses.

Table 6 presents the trading performance of the full model during the testing period. Compared to the baseline model, the full model demonstrates substantial improvements across all performance metrics. The total win rates exceed 86% for all stocks, with Evergreen and TSMC reaching 89.82% and 88.30%, respectively, indicating strong predictive stability.

In terms of long and short trades, both exhibit consistently high win rates. Long trade win rates generally exceed 87%, while short trade win rates remain above 78%, suggesting that the model effectively captures both upward and downward market movements.

Strategy returns are particularly notable, with Evergreen, MediaTek, and TSMC achieving 598.82%, 563.71%, and 356.86%, respectively, all significantly outperforming the buy-and-hold strategy. Corresponding excess returns are also substantially positive. The average return per trade ranges from 0.996% to 5.59%, indicating stable profitability at the trade level.

Furthermore, the profit factor is consistently well above 1, reaching as high as 29.53 for Evergreen, reflecting a strong imbalance between total profits and losses. Overall, the full model demonstrates superior performance in terms of return, stability, and risk efficiency, indicating strong potential for practical trading applications.

7.3 Performance Comparison: BIGRU vs. HFSLS-PSO-BIGRU

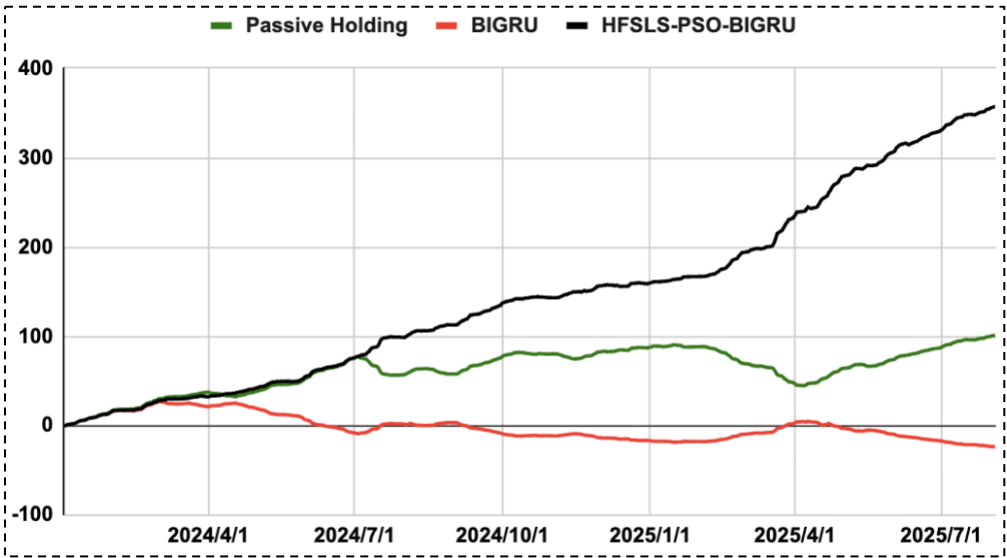


Fig 14. Cumulative return curves (TSMC)

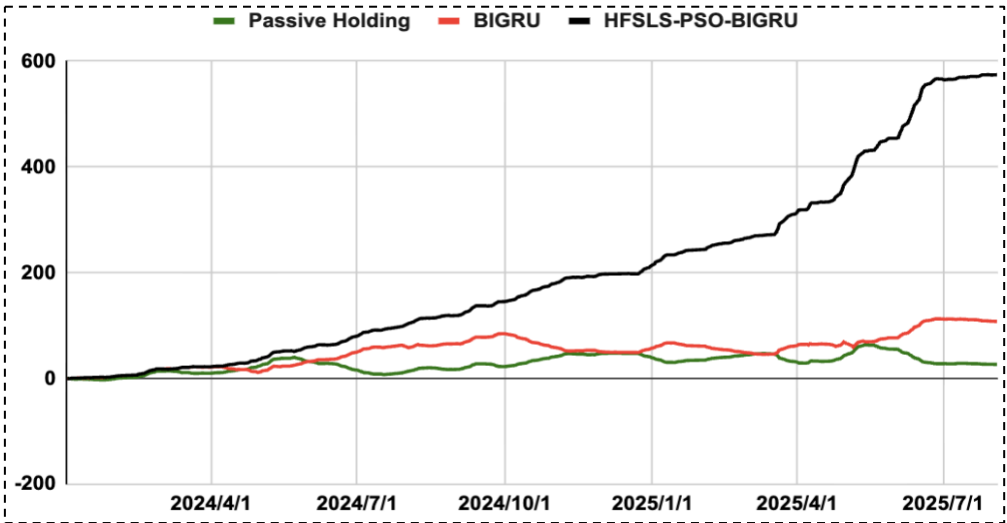


Fig 15. Cumulative return curves (Evergreen)

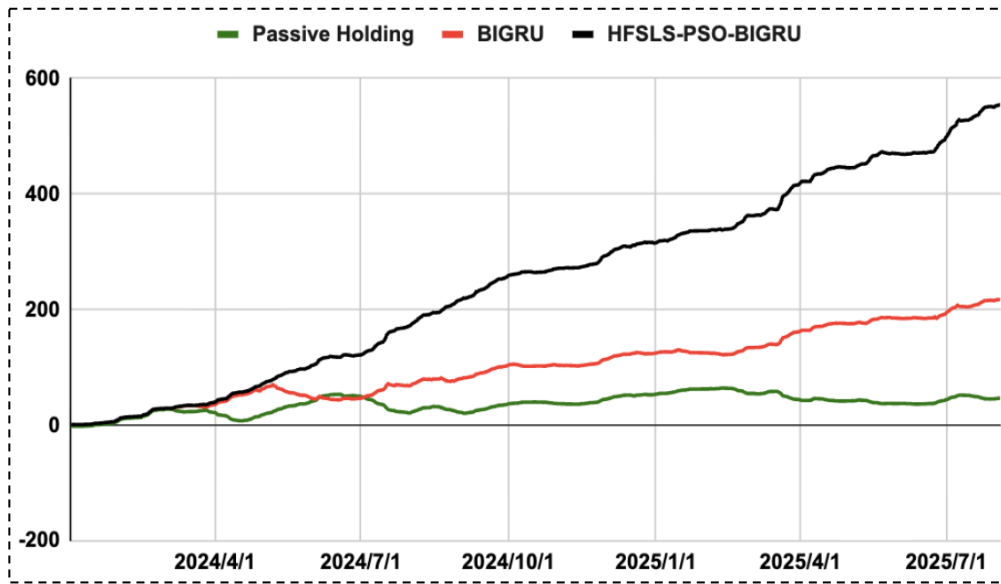


Fig 16. Cumulative return curves (MediaTek)

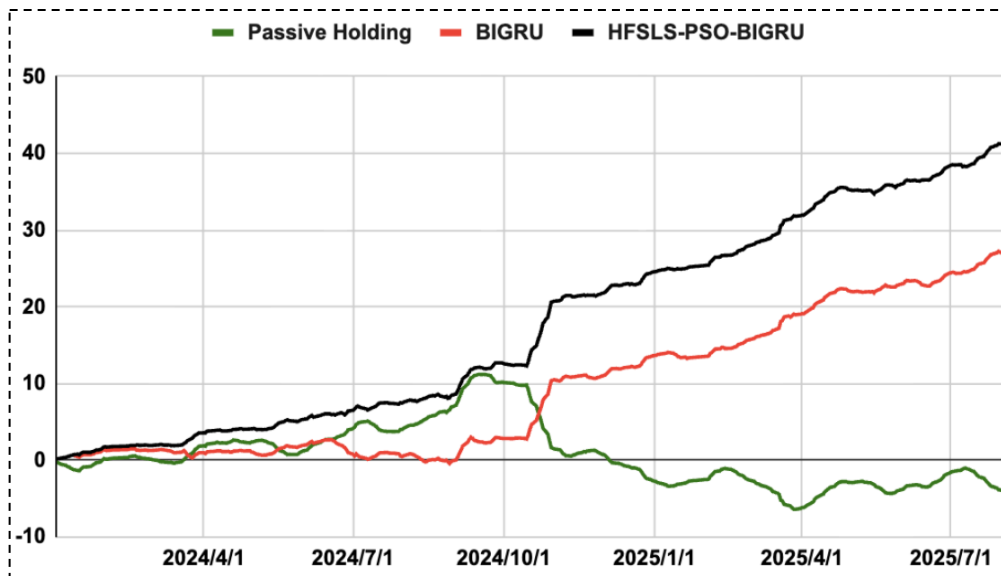


Fig 17. Cumulative return curves (Uni-President)

7.4 Cumulative Return Analysis: BIGRU vs. HFSLS-PSO-BIGRU

To evaluate the practical profitability of the proposed HFSLS-PSO-BIGRU framework, prediction signals were translated into a long-short trading strategy. The cumulative return curves were compared against the baseline BIGRU and the Passive Holding strategy (Fig 14–17).

As shown in Fig 14 (TSMC) and Fig 16 (MediaTek), the proposed model (black line) delivers exceptional profit momentum. While the passive holding strategy (green line) experienced stagnation or minor drawdowns in late 2024, our framework maintained a consistent upward trajectory, significantly outperforming all benchmarks.

The model's robustness is most evident in Fig 15 (Evergreen), a highly volatile shipping stock. Amid intense market fluctuations where both the baseline (red line) and passive holding struggled to generate meaningful returns, our proposed framework achieved exponential growth. By leveraging the HFSLS module to filter out stochastic noise, the model successfully captured primary trend reversals. Furthermore, in Fig 17 (Uni-President), the proposed model demonstrated superior downside protection; it effectively bypassed market downturns observed in late 2024, maintaining a positive return profile while other strategies suffered losses.

In conclusion, the proposed framework consistently yielded the highest cumulative returns across diverse stock profiles. These results confirm that the HFSLS-PSO-BIGRU framework provides superior risk-adjusted returns and excels at navigating nonlinear financial dynamics, underscoring its significant potential for real-world quantitative trading.

7.5 Comparative Analysis against TAIEX

Table 7. Performance Comparison of the HFSL-PSO-BIGRU Model Against the TAIEX

Performance Metric	TSMC	Evergreen	MediaTek	Uni-President
Total Trades	286	286	286	286
Long Trades	180	133	140	127
Short Trades	106	153	146	159
Long Win Rate (%)	76.667	83.459	80	30.709
Short Win Rate (%)	76.415	64.052	67.123	54.088
Total Win Rate (%)	76.573	73.077	73.427	43.706
Strategy Return (%)	203.137	304.597	289.862	30.829
TAIEX Return (%)	28.286	28.286	28.286	28.286
Excess Return (%)	174.851	276.311	261.576	2.543
Profit Factor	9.885	6.526	6.084	1.046

Note: Excess Return is calculated as the difference between the Strategy Return and the TAIEX

Return. Profit Factor is defined as the ratio of total profits to total losses.

To evaluate the relative performance of the proposed trading strategy, this study further compares the strategy's returns with the Taiwan Capitalization Weighted Stock Index (TAIEX). This comparison helps assess whether the strategy can generate excess returns and evaluate its performance relative to the overall market. It should be noted that the original test dataset contained 393 observations; however, when conducting the comparison with the market benchmark, the sample size was adjusted to 286. This reduction is primarily due to the absence of corresponding TAIEX data for certain trading dates. To ensure fairness and consistency in the comparison, only samples that contain both individual stock and TAIEX data were retained for analysis. This data filtering approach prevents bias caused by missing data and ensures that the comparison between strategy returns and the market benchmark is based on a consistent time frame.

According to the results in Table 7, for TSMC, Evergreen Marine, and MediaTek, the strategy returns significantly outperform the TAIEX, indicating superior relative performance for these stocks. In contrast, although Uni-President Enterprises' strategy returns also exceed the TAIEX, the magnitude of excess returns is relatively limited. Furthermore, data from Table 7 show that high-return stocks (such as Evergreen Marine and MediaTek) generally exhibit higher total win rates and profit factors, whereas

lower-return stocks (such as Uni-President Enterprises) have relatively lower profit factors.

7.6 Performance Comparison: HFSLS-PSO-BIGRU vs. TAIEX

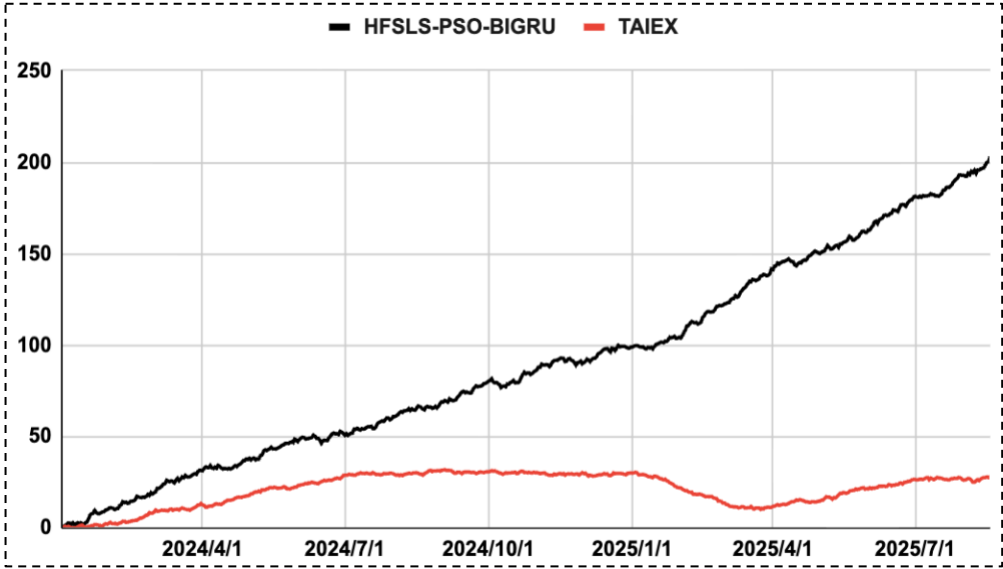


Fig 18. Cumulative return curves (TSMC)

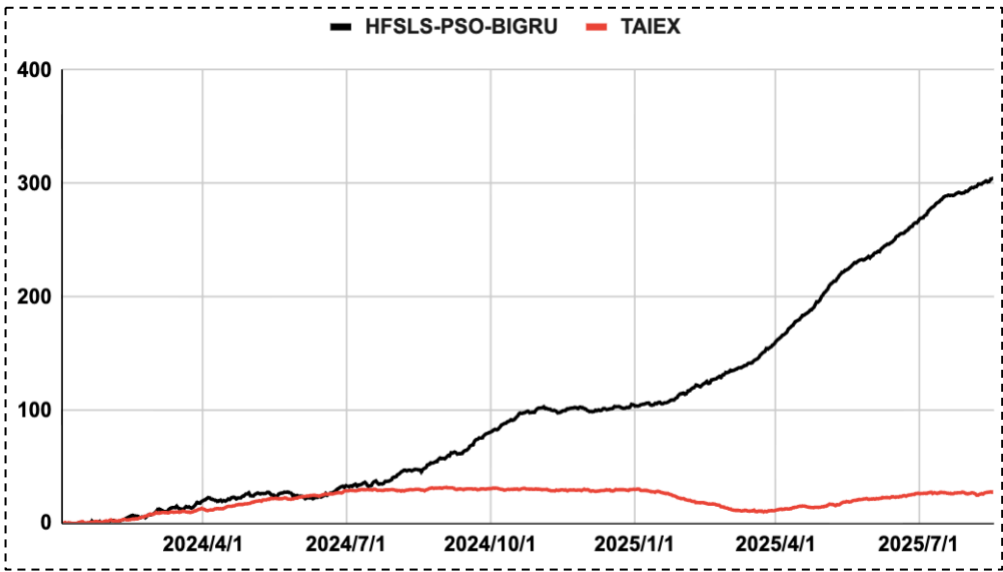


Fig 19. Cumulative return curves (Evergreen)

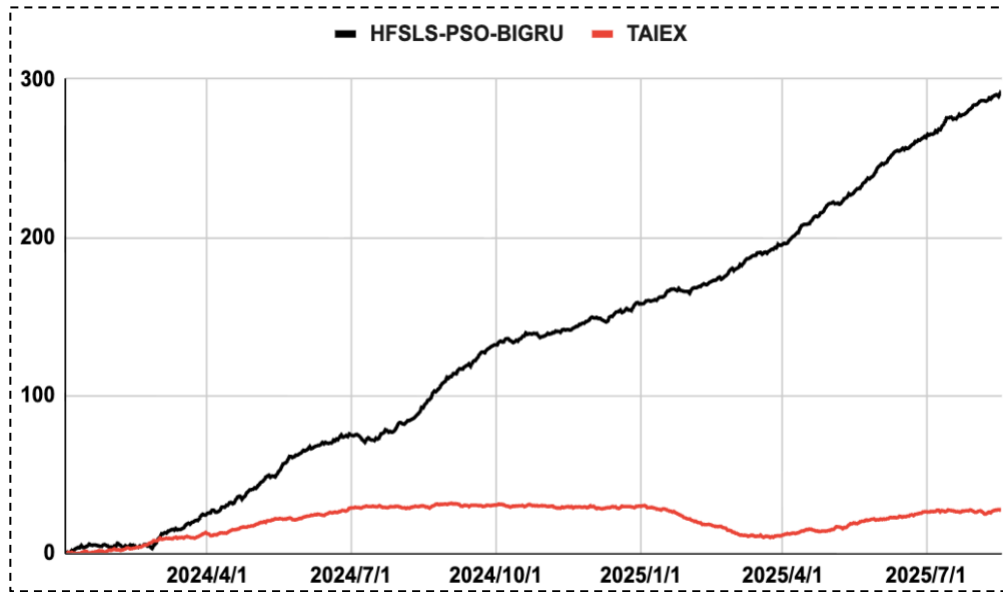


Fig 20. Cumulative return curves (MediaTek)

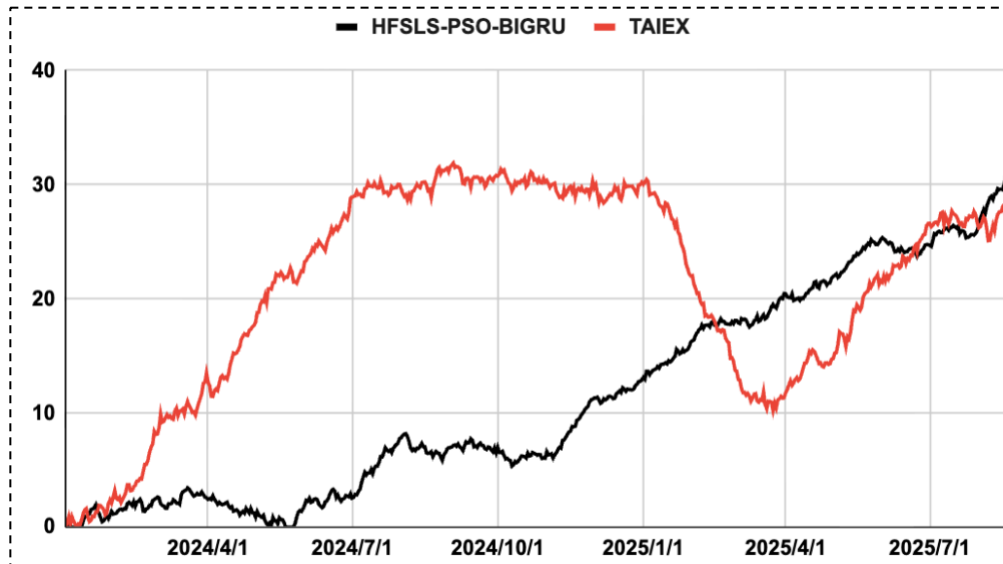


Fig 21. Cumulative return curves (Uni-President)

7.7 Cumulative Return Analysis: HFSLs-PSO-BIGRU vs. TAIEX

This section evaluates the empirical performance of the HFSLs-PSO-BIGRU framework using cumulative return curves (Fig 18–21), with the Taiwan Capitalization Weighted Stock Index (TAIEX) as the market benchmark. The results indicate that the proposed model can capture excess returns across stocks with different sector characteristics.

In Fig 18 (TSMC), 19 (Evergreen Marine), and 20 (MediaTek), the proposed model demonstrates consistently stronger upward performance compared to the benchmark. While the TAIEX (red line) exhibits relatively moderate growth over the observation period, the proposed framework (black line) maintains a steeper cumulative return trajectory. As a result, the performance gap between the model and the benchmark gradually widens over time.

Fig 21 (Uni-President) provides additional insight into the model's performance under more challenging conditions. Although the proposed framework does not consistently outperform the benchmark in terms of win rate for this stock, it still achieves higher cumulative returns over the full evaluation period. Notably, during the market downturn in early 2025, the benchmark experiences a significant drawdown, whereas the proposed model maintains a relatively stable performance path. This suggests improved downside resilience and more selective exposure to adverse market movements.

Overall, the results in Fig 18–21 suggest that the proposed framework performs well across both momentum-driven and defensive market conditions, demonstrating robust return generation capability relative to the benchmark.

8 Conclusion

This study proposes the HFSLS-PSO-BIGRU hybrid forecasting model, which integrates hierarchical feature selection, particle swarm optimization, and bidirectional gated recurrent units to address high dimensionality, nonlinearity, and long-term dependencies in financial time series. The model first selects key features and then optimizes parameters, significantly improving predictive accuracy and stability, demonstrating its feasibility in financial data analysis.

Regarding trading performance, the model consistently outperforms the market benchmark across most assets in the Taiwan stock market. Notably, for TSMC, Evergreen Marine, and MediaTek, the strategy achieves substantially higher returns than the Taiwan Capitalization Weighted Stock Index (TAIEX) and maintains stability during periods of market volatility, indicating its capacity to generate excess returns and adapt to risk.

In terms of robustness and applicability, the model is validated across different industry sectors, including technology, shipping, and consumer goods, maintaining stable predictive performance. Although performance varies among individual assets, the overall results demonstrate consistent and reliable effectiveness, reflecting the model's ability to handle diverse market conditions.

In summary, the study confirms that the HFSLS-PSO-BIGRU model effectively captures nonlinear patterns in financial markets, enhances predictive capability for investment strategies, and provides practical cross-industry applicability, offering empirical support and a methodological foundation for quantitative trading strategy development.

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